

D3 Path Difference

Calculate the gaps in the table. The speed of sound in air is 330 m s^{-1} .

	Wavelength	Path difference	Phase difference	Fully constructive interference (Y/N)	Fully destructive interference (Y/N)
D3.1	60 cm	90 cm	(a)	(b)	(c)

- D3.2 A 3.00 GHz microwave signal reaches a detector from a source by two routes. The shorter route is 4.500 m. If the two signals interfere destructively, what is the shortest distance possible for the other route?
- D3.3 Two speakers are set up 10.0 m apart in an auditorium, pointing at each other. A pure sound of frequency 440 Hz is being played through them. You may assume that the phase difference of the signals driving the speakers is 0° .
- a) A person is standing on the line joining the speakers, 0.30 m from the mid point. Calculate the phase difference as it would be detected by the person.
 - b) The person moves to the mid point between the speakers, where the sound is loudest due to constructive interference. They then walk towards one speaker passing the point where the sound waves cancel out and keep walking until the next loud point. How far do they walk from the mid point to the next loud point?

CHAPTER D. WAVES

Answers

D3.1 $\frac{\text{path difference}}{\lambda} = \frac{90}{60} = 1.5$

Removing the whole wavelength we get 0.5.

So the phase difference is $0.5 \times 360^\circ = 180^\circ$.

This is fully destructive.

D3.2 $\lambda = \frac{c}{f} = \frac{3 \times 10^8}{3 \times 10^9} = 0.100 \text{ m}$

Need to add half a wavelength, so we get $4.50 + 0.05 = 4.55 \text{ m}$.

D3.3

a) Path difference = $2 \times 0.3 = 0.60 \text{ m}$.

$$\lambda = \frac{c}{f} = \frac{330}{440} = 0.75 \text{ m}.$$

$$\text{Phase difference} = \frac{0.60}{0.75} \times 360^\circ = 288^\circ.$$

b) Need a path difference of $\lambda = 0.75 \text{ m}$.

$$\text{Distance moved} = \frac{0.75}{2} = 0.375 \text{ m}.$$